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Title: *Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study*

Journal: *Knowledge-Based Systems* (Elsevier)

To: Dr. Hang Yu, Senior Editor

Re: Response to Reviewers — Revision Round 1

Date: 19 October 2026

Dear Editor and Reviewers,

We thank you for your careful and constructive evaluation of our manuscript. Your comments prompted substantive revisions to the experimental account, statistical reporting, notation, and scope of our conclusions. We respond to each comment below and indicate the corresponding changes in the manuscript.

The most consequential correction concerns the original synthetic-exposure comparison. During revision, we identified that its baseline and intervention conditions had been evaluated through different pipeline paths, including differences in the construction of the baseline-correct question set (K_0). Consequently, we no longer rely on that comparison as evidence for the previously reported intervention effect. The revised manuscript instead presents paired dose-response experiments conducted under consistent pipelines. On Qwen, reducing synthetic exposure attenuated the Generation 5–10 retention decline at the 50% reduction level. On Gemma 3, the corresponding intervention improved Generation 10 retention but did not attenuate the decline over that interval. We have revised the interpretation to preserve this distinction rather than proposing a common mechanism that the experiments do not establish.

We have also clarified the status of effective training pressure (ETP). It is now presented as an operational framework for organizing experimental conditions and their observed diagnostics, not as a validated universal scalar predictor or an architecture-independent threshold. Additional revisions distinguish controlled comparisons from descriptive contrasts, report the available seed-level uncertainty, define the notation used in the equations and figures, and state the limits of the factual-retention evaluation.

The responses that follow identify the action taken for each concern and direct the reader to the relevant part of the revised manuscript. Revised manuscript passages quoted in this letter appear in [blue](#).

Respectfully submitted on behalf of all authors,

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Corresponding Author

Conventions used in this response

- Reviewer comments are reproduced *in green italics*, verbatim, including the table and section numbers of the *original* submission.
- Text newly added to the manuscript is quoted “[in blue, between quotation marks](#)”, so that every claim in this response can be located by text search in the revised manuscript.

- Locations are given as section and page of the revised manuscript. Where the revision renumbered a table or section, both numbers are given at first mention, e.g. Table 7 of the original submission (now Table 9, p. 27).
- All numerical values quoted here are generated from the archived per-seed results by `scripts/analysis/make_` and are reproduced verbatim from its output.

Experiment identifiers used in this response

The following identifiers appear in the revised manuscript and in this response. Table 3 of the revised manuscript maps the control experiments to the sources of variation they address.

ID	Experiment	Backbone	Seeds	Role
G1	QLoRA rank sweep, NF4	Qwen 2.5 1.5B	3 at $r=256$	Primary dose-response
G2	Pre-registered exposure dose-response, $r=10$	Gemma 3 1B IT	3 per dose	Cross-backbone replication
G3	Precision control: bf16	Qwen 2.5 1.5B	3 at $r=16$; 3 at $r=256$	Rules out quantization artefact
G4	Data-order control: three new shuffle seeds	Qwen 2.5 1.5B	3 at $r=256$	Rules out ordering artefact
G5	Pre-registered prospective rank $\times$ LR probe	Qwen 2.5 1.5B	1 per cell	Exploratory ordering test
B7	Pre-registered exposure dose-response, $r=256$	Qwen 2.5 1.5B	5 per dose	Primary exposure evidence
C1–C5	Exploratory intervention pilot (archived)	Qwen 2.5 1.5B	1	Superseded by B7; excluded from all estimates

Summary of changes

Forty review points were raised: three by the Editor, five by Reviewer 1, and thirty-two by Reviewer 2. Each is answered individually below. The table maps every point to the revision made and its location.

Point	Issue raised	Revision made	Location
Editor			
ED-1	Language editing	Complete language revision; terminology standardised throughout	Throughout
ED-2	Submission format	Seven documents in Word; figures at 300 dpi (non-PDF); <code>.bib</code> included	Submission package
ED-3	List of changes	This table plus an individual response to each point	This document
Reviewer 1			

Point	Issue raised	Revision made	Location
R1-0	Not a transferable predictor; single dataset; single-seed analyses; one exposure condition	Framework reframed as operational; scope and seed policy stated per claim	§3.5, §3.8, §6
R1-1	Prospective calibration PART	Five-step calibration protocol specified; held-out validation declared not performed	§3.5, §6
R1-2	Confound separation	Controls G3 (precision) and G4 (data order); fresh optimizer per generation; design/observed split	§3.9, Tab. 3, §4.6
R1-3	External validity PART	Strict exact-match re-scoring (Tab. S2); scope restriction stated	§3.2, §6, Tab. S2
R1-4	Exposure: Qwen versus Gemma 3	Pipeline inconsistency disclosed; two pre-registered dose–response experiments replace the pilot	§3.7, §4.5
R1-5	Reproducibility	Checkpoint revisions, prompts, seeds, masks, item-level vectors, raw generations and adapters deposited	§3.3, App. A
Reviewer 2			
R2-0	Framework conceptual; Gemma 4 uncited; limited scale	Contribution restated as design and evidence; Gemma 4 cited and disambiguated; scale stated as a limitation	§1, §3.3, §6
R2-1	Abbreviations and notation (10 points)	All acronyms expanded; notation table rebuilt; set/cardinality and singular-value notation corrected	Tab. 1, §3.2–3.6
R2-2	Malformed equations (4 points)	Glyph fault traced and corrected; Eq. 1–6 rewritten; glyph audit script included	Eq. 1–6, Tab. 9
R2-3	Internal inconsistencies (7 points)	Each corrected; threshold language unified; seed claims aligned	Abstract, §4, §5
R2-4	Methodology (6 points)	Inferential statistics added; per-backbone calibration proposed; “sharp” withdrawn; Gemma 4 sourced	§3.5, §4, §6
R2-5	References (5 points)	Duplicate arXiv resolved; year corrected; Gemma 4 added; preprint status declared; list harmonised	<code>cas-refs.bib</code>

Editor

Editor, Submission requirements — Format of the revised submission:

While submitting revision, please ensure to submit a list of changes or a rebuttal against each point that was being raised. The revised cover letter, response to reviewers, highlights, revised manuscript, credit author statement, author agreement and declaration of interest should all be in word format. Source file alone can be in word or LaTeX file and only one final version of manuscript is only required. The tables, figures and equations should be in editable format in the revised manuscript. Figures should be in good quality 300 dpi and not in pdf format. For LaTeX version, kindly ensure to check if supporting .bib files are also included.

Author response:

Each requirement is addressed as listed below.

Requirement	Compliance
(a) List of changes or rebuttal against each point	Both: a summary-of-changes table maps all forty review points, followed by an individual response to each.
(b) Seven documents in Word	Cover letter, response to reviewers, highlights, revised manuscript, CRediT author statement, author agreement and declaration of competing interest are submitted in .docx.
(c) Source file; one final version	A single final L ^A T _E X source is submitted with its compiled PDF.
(d) Editable tables, figures, equations	Tables are native <code>tabular</code> environments; equations are <code>amsmath</code> environments; no table or equation is an image.
(e) Figures at 300 dpi, not PDF	All figures are submitted at 300 dpi as PNG/TIFF. No figure is submitted in PDF.
(f) Supporting .bib	<code>cas-refs.bib</code> is included in the L ^A T _E X submission package.

Author action:

1. The seven required documents were produced in .docx.
2. All figures were regenerated at 300 dpi and exported to PNG, replacing the PDF versions used in the original submission.
3. `cas-refs.bib` was audited and is included in the submission package.

Revise (including language editing). Provide point-by-point response to each reviewer comment in the cover letter.

Author response:

Thank you for this guidance. We have edited the manuscript throughout for language and clarity, with particular attention to the description of experimental conditions, the distinction between results and interpretations, and the consistency of terminology across sections. We have also moderated claims that extended beyond the tested architectures or evaluation setting.

Author action:

The manuscript underwent a complete language revision prior to resubmission, covering grammar, terminology consistency, and the removal of residual non-English constructions. Terminology was standardised throughout (Homeostatic / Bounded / Degradative; “effective training pressure” written in full at every occurrence), and all abbreviations are now expanded at first use. Should the Editor consider a further round of professional language editing necessary, we will gladly commission Elsevier’s Author Services before acceptance. Every reviewer comment below is addressed by a dedicated revision, with the corresponding section and page cited.

Reviewer 1

Reviewer #1, General assessment — Scope of the evidence:

This paper studies recursive fine-tuning on synthetic data across Qwen 2.5 1.5B, Gemma 3 1B, and Gemma 4 E2B for ten generations. The systematic design, multi-seed headline experiments, and distributional diagnostics are valuable. However, the proposed pressure framework is not yet a transferable quantitative predictor, the main metric uses one short-answer dataset, several analyses are single-seed, and the exposure intervention is shown mainly for one Qwen condition. Major revision is required.

Author response:

We accept all four reservations and have restructured the paper so that each is stated as a property of the evidence rather than left for the reader to infer.

First, effective training pressure is no longer presented as a predictor. It is introduced as an operational experimental framework that organises three independently manipulated variables, and the operational index Π is labelled an exploratory post-training summary wherever it appears. Second, the single-dataset limitation is retained, but we add a strict exact-match re-scoring of all saved predictions to show that the regime classifications do not depend on the permissive matching criterion. Third, the seed policy is now explicit at the level of individual claims: headline conditions use three or five seeds, six for the pooled $r = 256$ estimate; intermediate ablations use one seed and are labelled descriptive; the two classes are never pooled. Fourth, the exposure axis is no longer shown for one Qwen condition: the revision replaces the single-seed pilot with two pre-registered dose–response experiments, one per backbone, each with multiple paired seeds at every dose level.

The one reservation we cannot remove is the first. Prospective calibration on an unseen backbone was not performed, and we now say so in those words.

Revised manuscript: §3.5, p. 8; §3.8, p. 9; §3.7 and §4.5, pp. 9 and 16–17; §6, pp. 21–22; supplementary Table S2.

Reviewer #1, Concern #1 — Prospective calibration of ETP:

How can effective training pressure be quantitatively calibrated and prospectively tested on an unseen backbone or dataset?

Author response:

We agree that a quantitative framework should not be presented as prospectively predictive without validation on held-out settings. The revised manuscript therefore distinguishes variables specified before training (nominal adapter rank, learning rate, planned training steps, and synthetic exposure) from quantities measured afterward (effective rank and weight drift). Retention, response length, and SDI-3 are evaluated as outcomes or diagnostics, not used to define a prospective threshold.

Within the tested protocols, we can locate changes in retention trajectories between adjacent configurations. Those intervals do not establish a universal numerical ETP threshold: the attempted cross-architecture normalisations did not align the observed boundaries. A prospective test on an unseen backbone or dataset would require a calibration rule and outcome criterion specified in advance, followed by evaluation on held-out conditions without refitting the rule to their results. We have added this as a proposed validation protocol, not as an experiment already completed.

The exploratory G5 experiment provides a narrower check: two of three pre-specified, single-seed configurations followed the expected ordering. No threshold was pre-specified for G5, and it did not evaluate an unseen backbone or dataset. We therefore describe ETP as an operational experimental framework rather than a validated, transferable predictor.

Author action:

We revised the ETP definition, qualified the G5 interpretation, and added the requirements for future prospective calibration.

Revised manuscript: Section 3.5 (ETP definition); Section 4.6 (G5 prospective test); Section 6 (prospective validation and limitations).

Reviewer #1, Concern #2 — Separating pressure effects from confounders:

How are quantization, optimizer state, data order, and model-specific dynamics separated from the claimed pressure effects?

Author response:

The revised manuscript identifies what each control can establish and what remains confounded. For quantization, G3 compares Qwen at $r = 16$ and $r = 256$ under the same bf16 protocol and three paired seeds. The per-seed Generation 10 differences are +15, +13, and +15 questions out of 78, a mean difference of +18.4 percentage points. The exact paired sign-flip test gives $p = 0.25$. This directionally consistent, small-sample result supports the rank contrast under bf16 but does not establish a statistically conclusive or architecture-independent effect.

By contrast, the 17.7-percentage-point Qwen difference obtained by combining conditions from different protocols is descriptive: rank, quantization, and aspects of the data-order protocol differ across the groups. We do not use that comparison as a rank-only hypothesis test. Comparing NF4 and bf16 provides an additional implementation check, not a complete causal account of the observed trajectories.

A fresh optimizer is instantiated for each generation, so optimizer state is not carried across generations under the reported protocol. In G4, both the random seed and data order changed

relative to G1. The Generation 10 comparison ($N = 3 + 3$, permutation $p = 0.70$) consequently does not isolate an order effect; nor does the nonsignificant result prove that order has no effect.

Model-specific dynamics likewise remain relevant. For Gemma 3, the $r = 4$ versus $r = 16$ contrast involves five runs per condition. An independent-seed exact permutation test yields $p = 0.0079$, whereas a paired sign-flip analysis yields $p = 0.0625$. We report both results with their differing seed-assignment assumptions rather than treating either test as universally applicable.

Author action:

We separated design variables from post-training diagnostics, documented the controls and their limitations, and distinguished the paired G3 comparison from the descriptive cross-protocol Qwen contrast.

Revised manuscript: Sections 3.5 and 3.9 (variables and controls); Section 4.1 (rank sweep); Section 4.2.1 (Qwen comparisons); Section 4.3.1 (Gemma 3); Section 4.6 (G3 and G4).

Reviewer #1, Concern #3 — External validity of the evaluation:

Do the conclusions hold for longer answers, multi-hop questions, other domains, and larger factual evaluation sets?

Author response:

Our results do not establish generalization to those settings. The evaluation concerns short-answer factual questions. It uses a held-out set of 200 questions disjoint from the 2,000-question synthetic training stream; retention is calculated over questions answered correctly at generation 0 ($|K_0| = 78$ for Qwen under the reported protocol).

The primary scorer permits bidirectional substring matches to accommodate short-answer variants. Because longer generated answers could benefit from that permissive criterion, we also report a strict exact-match re-evaluation of saved predictions. Absolute scores are lower under the strict scorer, while the reported qualitative trends for the evaluated conditions are retained. This sensitivity check addresses one scoring concern; it is not a test of long-form generation or multi-hop reasoning.

We have corrected the discussion of evaluation-set size. Expanding a held-out, disjoint evaluation set would not, by itself, change the composition of the training stream. Larger factual test sets would improve the assessment of sampling variability and generality, while other domains and answer formats require separate experiments.

Author action:

We clarified the scorer and the training–evaluation separation, added the strict-scoring comparison, and explicitly limited the conclusions to the tested short-answer factual setting.

Revised manuscript: Section 3.1 (evaluation protocol); supplementary Table S2 (strict scoring); Section 6 (scope and limitations).

Reviewer #1, Concern #4 — Exposure reduction and model-dependent responses:

Why does the five-percent exposure reduction stabilize Qwen but not Gemma 3, and does this generalize across ranks and seeds?

Author response:

This question exposed a substantive problem with the original comparison. C1 and C5 followed different evaluation pipelines, including differences in K_0 construction and data loading. The previously reported 9.0- and 9.4-percentage-point intervention gains cannot therefore serve as evidence of stabilization. We also clarified that the pilot is excluded from all estimates; every dose in the two pre-registered experiments is specified as a fraction of training examples removed, with no token-based figure. Under a matched pipeline with three paired seeds, the estimated effect of that smaller intervention was +1.9 percentage points and was inconclusive.

The revised Qwen evidence comes from the B7 dose-response experiment: 0%, 10%, 25%, and 50% example removal at $r = 256$, with five paired seeds per dose. Relative to no removal, Generation 10 retention increased by +1.8 percentage points at 10% removal ($p = 0.160$), +4.9 points at 25% ($p = 0.005$, 95% CI [2.5, 7.2]), and +9.2 points at 50% ($p < 0.001$, 95% CI [6.9, 11.6]). At 50% removal, the Generation 5–10 retention slope was -0.15 percentage points per generation, compared with -1.13 without removal. Thus, attenuation of the later decline is supported for this Qwen configuration and protocol; it is not inferred from the invalid C1/C5 comparison.

Gemma 3 showed a different response. In G2, 50% removal at $r = 10$ increased Generation 10 retention by 7.7 percentage points across three paired seeds, but the Generation 5–10 slope was approximately -0.91 percentage points per generation in both arms. The intervention improved endpoint retention without demonstrably slowing later loss. Neither experiment identifies the mechanism or establishes that the effect generalizes across untested ranks. Similar point estimates for 50% exposure removal and a halved learning rate are not presented as formal equivalence.

Author action:

We disclosed the pipeline inconsistency, withdrew the original C1/C5 inference, replaced it with the paired B7 dose-response analysis, and reported the distinct Gemma 3 outcome without attributing both patterns to the same mechanism.

Revised manuscript: Section 4.5 (dose-response experiments B7 and G2); Section 4.6 (learning-rate comparison); Section 6 (mechanistic and generalization limits).

Reviewer #1, Concern #5 — Scope of the reproducibility package:

What complete code, checkpoints, prompts, seeds, and raw outputs will support independent reproduction?

Author response:

We agree that aggregate tables alone are insufficient and that the scope of the public materials must be described precisely. The revised reproducibility description identifies the model checkpoints, question partitions and baseline-correct K_0 sets, generation and evaluation prompts, decoding settings, training configurations, seeds, and the per-seed results used in the reported analyses. It also directs readers to the associated archive at <https://doi.org/10.5281/zenodo.23146342>.

Importantly, we distinguish archived analysis materials from artifacts that are not publicly deposited. The described release includes consolidated results and replication scripts, but raw token-level predictions, trained adapter checkpoints, and base-model weights are not included. We

therefore do not claim that the archive alone permits a complete token-level audit or byte-for-byte reconstruction of every training run. Scripts introduced or revised after release v1.0.3 must be identified by the version in which they are actually deposited; they should not be attributed to the earlier release.

Author action:

We expanded the reproducibility and data-availability description to state what the public package contains, what it does not contain, and which archive version supports each reproduction claim.

Revised manuscript: Appendix A.1 (reproducibility and data availability).

Reviewer 2

Reviewer #2, General assessment — Status of the framework and of the cross-architecture claims:

The paper addresses a timely and practically relevant question and provides a useful empirical dose-response characterization. However, the current manuscript has major issues in notation, equations, table/figure formatting, internal consistency, statistical rigor, and citation accuracy. The “effective training pressure” framework is promising but currently conceptual rather than quantitative. The cross-architecture claims are weakened by the uncited Gemma 4 E2B model and limited scale. I recommend major revision before further consideration.

Author response:

We agree with the characterisation and have treated the two substantive objections as corrections rather than as matters of emphasis.

On the status of the framework: the reviewer is right that effective training pressure, as submitted, was a conceptual label for three already-known axes and not a quantitative measure. We do not defend the original framing. The revised manuscript states that no scalar predictor is derived, separates the variables fixed before training from the diagnostics measured after it, and restates the contribution as the factorial design, the dose-response evidence along three axes, the two pre-registered exposure experiments, and the finding that the regime boundary is backbone-dependent and not reconciled by any architectural normalisation we tested.

On the cross-architecture claims: Gemma 4 E2B IT is now cited, explicitly distinguished from Gemma 3n E2B, and its architectural description is attributed to the released model configuration. Because that backbone was evaluated with a single seed, it is labelled supportive validation and excluded from every threshold estimate. The restriction to 1–2 B parameter models is retained as the first limitation in Section 6.

Beyond the points listed below, the reviewer’s objection to the word “confirms” in one place prompted us to apply the same standard everywhere. We re-read the Results and Discussion for claims whose strength exceeded the design supporting them, and corrected fifteen further instances: a single-seed grid described as *demonstrating* joint dependence, now reported as descriptively consistent with it; a degradative regime described as *reversible*, now described as attenuated to the bounded regime on one backbone and as a retention gain without slope reduction on the other; a 50 % exposure reduction described as *arresting* progressive loss, now quantified as reducing its rate to -0.15 pp/generation; a single-seed observation on a third backbone described as *generalising*,

now reported as consistent with but not establishing generalisation; and a non-monotonic slope response across doses that the original text had smoothed, now reported as two distinct outcomes. We also found one paragraph in Section 5.4 that still drew a conclusion from the archived pilot described in our response to Reviewer 1, Concern #4, and removed it.

Revised manuscript: §1, p. 3; §3.3, p. 6; §3.5, p. 8; §4.1, p. 10; §4.3.2, p. 14; §5.1–5.4, pp. 18–20; §6, pp. 21–23.

Reviewer #2, Concern #1 — Abbreviations and notation:

ETP appears in Figure 1 but is never explicitly defined as “effective training pressure.” MTLD is introduced without expansion. KL, JS, and coverage appear in Figure 1 without definition. bf16, SVD, TCE, QLoRA, and LoRA are used without consistently expanded forms. “Homeo.” is used in Table 2 without definition. K_0 is used both as a set of questions and as a retention metric. T , $D1_T$, $D10$, ΔI , $\text{Instability}(1)$, MeanLen_T , and related symbols in Eq. 6 are not fully defined. The effective-rank formula uses σ_i as both singular values and normalized singular values. The normalization condition is mathematically incorrect as written. BA_F , $\theta_i^{(t)}$, $\theta_i^{(0)}$, and related notation in the drift equation are not consistently defined. $r \in \{4, 10, 12, 14, 16\}$ set notation was garbled in the original PDF.

Author response:

We agree that the original notation was insufficiently self-contained. The revised manuscript introduces effective training pressure in full before its use in Figure 1 and expands retained abbreviations at first occurrence, including Measure of Textual Lexical Diversity (MTLD), 16-bit brain floating point (bf16), singular value decomposition (SVD), Truncated Cross-Entropy (TCE), Low-Rank Adaptation (LoRA), and Quantized Low-Rank Adaptation (QLoRA). KL, JS, and coverage have instead been removed from Figure 1 because they are not computed in the reported experiments. “Homeo.” has been replaced with “Homeostatic.”

We have also separated the set from the metric: K_0 denotes the set of held-out questions answered correctly at generation 0, $|K_0|$ its cardinality, and $R(t)$ the retention ratio at generation t . For SDI-3, T denotes the evaluation horizon, $D_{1,t}$ denotes Distinct-1 at generation t , and $D_{1,0}$ denotes its generation-0 value; the comma distinguishes the latter from the ambiguous D_{10} . We define $\Delta I = \text{Instability}(T) - \text{Instability}(1)$, where $\text{Instability}(t)$ is the fraction of training items whose normalized response changes between generations $t - 1$ and t . MeanLen_t denotes mean response length at generation t .

Author action:

We revised Figure 1, the notation table, first-use acronym definitions, the regime-table labels, and the text preceding Eq. 6.

Revised manuscript: Figure 1 and caption; notation and regime tables; Section 3.5; definitions surrounding Eqs. 1 and 6.

Reviewer #2, Concern #2 — Equations and tables:

Equations 1-6 are severely malformed in the provided PDF. For example, Eq. 1 uses “Retention.t /” instead of a clear function notation and has garbled summation over K_0 . Eq. 2 for effective rank is missing the negative sign and summation. It should be something like $\exp(-\sum p_i \log p_i)$, with $p_i = \sigma_i / \sum_j \sigma_j$. Eq. 6 for SDI-3 is unclear: signs, log ratios, and the ΔI term are not consistently defined. What is 5×10^6 in Table 7? Please check also elsewhere.

Author response:

Two of these points are defects in the mathematics and two are defects in the rendering, and we treat them separately because they have different causes.

The rendering fault. Rather than re-render the document and hope for a different outcome, we traced it. Two audits establish where it does *not* originate. First, an automated scan of the complete L^AT_EX source returns a single non-ASCII character, inside a comment: no mathematical symbol is encoded as raw Unicode, and the source is syntactically correct as submitted. Second, `pdffonts` confirms that every font in our locally compiled PDF is embedded with a complete ToUnicode mapping, and text extraction from that file reproduces the equations correctly. The corruption is introduced when the uploaded file is re-processed: the embedded fonts use a custom encoding, and the re-distillation step does not preserve their Unicode mappings, so that minus signs, multiplication signs, set-membership symbols and percent signs are drawn correctly on the page but map to unrelated code points on extraction. This accounts for the specific form of the corruption the reviewer observed (“Retention.t /”, “5110^6”, “r E 4,10,12,14,16”) rather than missing or blank glyphs. For this revision the font pipeline no longer relies on custom-encoded subsets. Following the reviewer’s instruction to check elsewhere, we also wrote an audit that compares the extracted text of every equation, table cell, figure label and caption against the L^AT_EX source; its script and output are included in the replication package.

The mathematical defects. Equation 2 was wrong as written, not merely mis-rendered, and we have adopted the reviewer’s formulation. The original text used σ_i for both the singular values and their normalisation, which makes the normalisation condition circular. The revision introduces a separate symbol for the normalised weights and states the condition explicitly. Equation 6 was underspecified: the sign convention and the direction of the diversity ratio were not stated, so the sign of SDI-3 could not be interpreted. All three terms are now defined, and the direction of the composite is stated.

Point raised	Correction
(a) Eq. 1 notation and summation	Rewritten with explicit function notation and set cardinality: $R(t) = \frac{ \{ q \in K_0 : q \text{ answered correctly at generation } t \} }{ K_0 }$ The numerator is the cardinality of a set, not a summation, which is what the garbled original obscured. (§3.2, p. 5)

(b) Eq. 2 missing sign and summation; normalisation incorrect	Accepted as a mathematical error and corrected in the notation the reviewer proposed: $\text{erank}(BA) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}$ with “$\hat{\sigma}_i \geq 0$ and $\sum_i \hat{\sigma}_i = 1$” stated in Table 1. The symbol $\hat{\sigma}_i$ denotes the normalised singular weight; σ_i are the raw singular values of ΔW. (Eq. 2, §3.5.1, p. 8; Table 1, p. 25)
(c) Eq. 6 signs, log ratios, ΔI	Rewritten with each ratio oriented so that all three terms increase under deterioration: $\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I$ with $\Delta I = \text{Instability}(T) - \text{Instability}(1)$, the convention $0 \log 0 = 0$, and the interpretive statement “a positive SDI-3 score indicates cumulative distributional deterioration”. Each term is named in the preceding prose: verbosity increase, unigram diversity decrease, loss of item-level response stability. (Eq. 6, §3.6, p. 9)
(d) 5×10^{-6} ; check elsewhere	The value is the learning rate 5×10^{-6} in Table 7 of the original submission (now Table 9, p. 27); both the multiplication sign and the negative exponent were lost to the glyph fault. The instruction to check elsewhere was followed systematically: the audit covers every equation, table cell, figure label and caption, and also recovered the garbled set expressions, now rendered as $r \in \{4, 10, 12, 14, 16\}$ and $r \in \{10, 12, 14\}$. (Table 9, p. 27; §3.4, p. 7)

Author action:

1. The font pipeline was changed so that no embedded subset uses a custom encoding, and the result was verified at the point where the fault appears rather than only locally.
2. Equations 1, 2, 3 and 6 were rewritten and cross-checked against Table 1 term by term; Equation 3 now shows the absolute value and the summation explicitly.
3. An automated glyph audit was written and is included in the replication package, with its output, so that the check can be repeated by a reader.
4. Every figure was regenerated from source at 300 dpi, and the numerical values plotted in each were re-verified against the corresponding table, so that no figure reports a value different from the table it summarises.

Revised manuscript: §3.2, p. 5; §3.4, p. 7; §3.5.1, p. 8; §3.6, p. 9; Table 1, p. 25; Table 9, p. 27; replication package (audit script and output).

Reviewer #2, Concern #3 — Technical and data inconsistencies:

(a) *Qwen threshold inconsistency: Abstract and Section 4.2.3 say the boundary is between effective ranks 50–88; Section 5.1 says Qwen remains outside the degradative regime “up to effective rank 150.” This contradicts Table 2, where $r = 256$ has effective rank 87.57 and is degradative. (b) $r = 128$ is classified as “Bounded” in Table 2, but Section 4.4.3 calls $r = 128$ and $r = 256$ “above-threshold configurations.” (c) $r = 256$ is repeatedly called a “boundary configuration”, but Table 2 classifies it as degradative. (d) *Gemma 3 degradative range: Section 5.2 says “ $r \geq 16$ on Gemma 3,” but Table 4 shows $r = 10, 12, 14$ already degradative at Gen 5. (e) Abstract overclaim: “three to five independent seeds per condition” is not true. (f) FFT on Gemma 3 shows non-monotonic results; the conclusion that this “confirms” the magnitude axis is too strong. (g) C5 removal percentage: Section 3.4 mentions “15% removal,” while Section 4.5 reports approximately 5% token reduction.**

Author response:

All seven are accepted. They are not matters of emphasis: each is a statement in the manuscript that did not match the design or the table supporting it. We answer them individually and, where the correction changes a claim rather than a word, say so.

(a) *Qwen threshold stated inconsistently.*

The phrase “up to effective rank 150” was wrong and has been removed. The reviewer’s reading of Table 4 is correct: $r = 256$ has mean effective rank 87.57 and is degradative, so no claim of homeostasis beyond that value can stand. The deeper problem the reviewer exposes is that we were using a single figure to describe two different boundaries. The revision distinguishes them. On Qwen, the tested grid places the transition from homeostatic to bounded behaviour between effective ranks 29.5 ($r = 64$) and 50.2 ($r = 128$), and the transition from bounded to degradative between 50.2 and 87.6 ($r = 256$). The “50–88” interval quoted in the Abstract is the second of these. On Gemma 3, where no bounded regime was observed, the “3–6” interval is the transition from homeostatic directly to degradative. The two intervals are therefore not the same kind of object, and the revision no longer compares them as though they were. “(§4.2.3, p. 14; §5.1, p. 18; Abstract, p. 1)”

(b) $r = 128$ labelled above-threshold while classified Bounded.

Corrected. “Above-threshold configurations” now refers exclusively to $r = 256$. The $r = 128$ configuration is described as a “transition-zone configuration”, in which distributional signatures emerge without a progressive retention trajectory, which is precisely the dissociation we report for it, and which the original label obscured. “(§4.4.3, p. 15; §5.3, p. 19)”

(c) $r = 256$ called a boundary configuration.

Corrected throughout. The reviewer is right that the boundary lies between $r = 128$ and $r = 256$, not at $r = 256$. Every occurrence now reads “lowest-pressure degradative configuration tested”, which states what the configuration is without implying that it sits on the boundary. “(§3.4, p. 9; §4.5, pp. 16–17; §5.4, p. 20)”

(d) *Gemma 3 degradative range.*

Corrected. Table 4 of the original submission (now Table 6, p. 27) reports 78.3%, 73.9% and 69.6% at $r = 10, 12$ and 14 respectively, all degradative at Gen 5, so “ $r \geq 16$ ” was inconsistent with our own data. The text now reads “ $r \geq 10$ on Gemma 3” and locates the boundary “below $r = 10$ ”, between effective ranks 3 and 6. “(§5.2, p. 19; Table 6, p. 27)”

(e) *Seed count in the Abstract.*

Corrected, and the correction goes beyond the Abstract. The seed policy is now stated at the level of individual claims rather than once in aggregate: headline conditions use three or five seeds (six for the pooled $r = 256$ estimate), intermediate ablations use a single seed and are labelled descriptive, and the two classes are never pooled when a headline claim is made. An N column was added to the rank-sweep and dose-response tables; the remaining tables (Tables 7–9) state N in a footnote. Contribution 1 in the Introduction, which carried the same overclaim, was corrected with it. “(Abstract, p. 1; §1, p. 3; §3.8, p. 9; Tables 4–10)”

(f) *FFT on Gemma 3 described as confirmatory.*

Accepted. A single-seed, non-monotonic result cannot confirm anything, and “confirms” has been replaced by “is consistent with”. The non-monotonicity is now reported rather than smoothed over: Gen 5 retention was 78.3% at $\eta = 10^{-6}$, 54.3% at 5×10^{-6} , and 63.0% at 10^{-5} , and the text states that “the non-monotonic pattern at $\eta = 5 \times 10^{-6}$ limits interpretation. We treat this as descriptive evidence rather than confirmatory generalization.” Contribution 2 in the Introduction carried the same overclaim and was corrected. “(§1, p. 3; §4.3.3, pp. 14–15)”

(g) *C5 removal percentage.*

The reviewer identified a genuine ambiguity, and the revision removes it rather than reconciling it. Two things were conflated: the fraction of training *examples* removed and the resulting reduction in *tokens*, which differ because responses vary in length. More substantively, the comparison those figures supported is no longer used: the C1–C5 pilot was found to compare arms produced by different evaluation pipelines and is excluded from every estimate in the paper (see Reviewer 1, Concern #4). In the two pre-registered experiments that replace it, exposure is specified exclusively as the fraction of training examples removed — 0%, 10%, 25%, 50% — with no token-based figure anywhere. The pilot, including its original dose description, is archived in the replication package. “(§3.7, p. 9; §4.5, pp. 16–17; Figure 9, p. 25)”

Author action:

1. The phrase “up to effective rank 150” was deleted, and the two distinct boundaries are now named separately wherever a threshold is quoted.
2. The regime labels for $r = 128$ and $r = 256$ were corrected at every occurrence, by systematic search rather than selectively.
3. The Gemma 3 degradative range was corrected in the Discussion to match Table 6.
4. The seed claim was corrected in the Abstract and in Contribution 1, and an N column was added to Tables 4 through 10.
5. “Confirms” was replaced in Contribution 2 and in §4.3.3, and the non-monotonic values are now reported explicitly.
6. The example/token ambiguity was removed by restricting all dose specifications to fractions of training examples.
7. Beyond the seven points, we re-read the Results and Discussion for statements of the same kind and corrected three further instances in which a horizon (Gen 5 versus Gen 10) or a seed count was implied but not stated.

Revised manuscript: Abstract, p. 1; §1, p. 3; §3.4, p. 9; §3.7, p. 9; §3.8, p. 9; §4.2.3, p. 14; §4.3.3, pp. 14–15; §4.4.3, p. 15; §4.5, pp. 16–17; §5.1–5.4, pp. 18–20; Tables 4–10, pp. 26–27.

Reviewer #2, Concern #4 — Methodology:

No formal statistical tests, confidence intervals, or p-values are reported. The effective training pressure framework is not quantified. Cross-architecture normalization failed, but no alternative calibration method is proposed. “Effective training pressure” is not a new quantitative measure; it is a conceptual label for three already-known axes. The Gemma 4 E2B backbone is a serious concern: it lacks a citation, may be confused with Gemma 3n E2B, and its architecture is described without a source. The “sharp transition” claim is weakened by sparse rank grids and wide threshold intervals.

Author response:

(a) *No formal tests, confidence intervals, or p-values.*

Inferential statistics have been added wherever the design admits them, and the design is now stated alongside each test rather than left to be inferred. For the Gemma 3 anchor-rank contrast ($N = 5$ per rank, $|K_0| = 46$), the mean difference is +37.8 pp; because the archived data do not confirm whether seeds were paired across ranks, we report both possibilities — an exact sign-flip test at $p = 0.0625$ if paired, a bilateral permutation test over $\binom{10}{5} = 252$ allocations at $p = 0.0079$ if independent — with Hedges’ $g \approx 13.3$ under either reading. For the Qwen rank contrast, the clean comparison is the G3 paired design (same seeds, same precision, rank the only variable): per-seed differences of +15, +13 and +15 items out of 78, mean +18.4 pp, exact sign test $p = 0.25$ at $N = 3$. We report that p -value rather than suppress it: the direction is consistent across all three pairs, but three pairs cannot produce a small p , and we say so. For the two dose–response experiments, paired t -tests with 95 % confidence intervals are reported at every dose level. Where groups are not exchangeable under the relevant null — the 17.7 pp gap between the three-seed $r = 16$ condition and the six-seed $r = 256$ estimate, which differ in rank, precision protocol and seed assignment simultaneously — the comparison is labelled descriptive and no test is reported. “(§4.1, p. 10; §4.2.1, p. 12; §4.5, pp. 16–17; §4.6, p. 17)”

(b) *The framework is not quantified.*

Correct, and the revision no longer implies otherwise. No scalar predictor is derived. The manuscript now separates the variables fixed before training — nominal rank, learning rate, training steps, dose fraction — from the quantities measured after it — effective rank, weight drift, response length, SDI-3 — and states that the latter “are not available prospectively; they therefore serve as post-hoc explanatory variables rather than pre-specified predictors”. The operational index $\Pi = \text{erank}(\Delta W) \cdot (\eta/10^{-5})$ is labelled “an exploratory post-training summary” and is “not used as a validated quantitative threshold”. Developing a quantitative formulation is listed as future work, with the protocol a prospective test would require specified in Section 6. “(§3.5, p. 8; §6, p. 22)”

(c) *Normalisation failed with no alternative proposed.*

The reviewer asks for the constructive half of a negative result, and the original manuscript supplied only the negative half. Five normalisations were tested — by hidden dimension, its square root, layer count, attention heads, and key–value heads — and none aligned the thresholds across backbones (Table 13, p. 28). We retain that table as a reported negative result and now propose what we believe the finding implies: that a transferable constant should not be sought, and that calibration must be per-backbone. Section 6 specifies the procedure concretely — a pilot sweep of two ranks spanning roughly an order of magnitude, five generations, single seed; the boundary

estimated as the interval between the highest configuration whose early retention slope is indistinguishable from zero and the lowest whose slope is negative — together with its cost, which is approximately two short runs per backbone. We present this as the practical alternative to a universal normalisation rather than as a validated method, since we have not executed it on an unseen backbone. “(§4.2.3, p. 14; §6, p. 22; Table 13, p. 28)”

(d) *Not a new quantitative measure.*

We agree, and we do not defend the original framing. Effective training pressure, as submitted, was a conceptual label for three axes already known individually, and presenting it as a measure was an overstatement. The revision restates the contribution as what the experiments actually establish: a factorial design over three axes; dose–response evidence along each; a retention/distribution dissociation at intermediate capacity that is not captured by existing accounts of recursive collapse; two pre-registered exposure experiments, one per backbone; and the finding that the regime boundary is backbone-dependent by roughly an order of magnitude and is not reconciled by any architectural normalisation we tested. The organising value of the framework is that three independently manipulated variables move the system across the same qualitative boundary, which is an empirical observation and is presented as one. “(§1, p. 3; §3.5, p. 8; §5.1, p. 18; Table 11, p. 28)”

(e) *Gemma 4 E2B: citation, possible confusion with Gemma 3n E2B, unsourced architecture.*

All three parts are addressed. The backbone is now cited as Gemma Team [41] at every point of introduction and description. On the possible confusion, which we should have pre-empted: the model is Gemma 4 E2B IT, a distinct release from Gemma 3n E2B, and the revision states this explicitly rather than relying on the reader to infer it from the version number. The architectural description is attributed to the released model configuration rather than asserted — 35 layers, hidden size 1536, grouped-query attention with 8 attention heads and 1 key–value head, and key–value projections shared across the final 20 of 35 layers, which is why a LoRA adapter targeting `q_proj` and `v_proj` adapts 35 query and 15 value matrices. The repository identifier and revision hash are listed in supplementary Table S1. We also constrain what the backbone is used for: it was evaluated with a single seed, so it is labelled supportive validation, excluded from every threshold estimate, and the limitation is stated in those terms — “they do not constitute an independent replication of the quantitative threshold estimates”. “(§3.3, p. 6; §4.2.2, p. 12; §6, p. 23; Table S1)”

(f) *“Sharp transition” weakened by sparse grids and wide intervals.*

Accepted, and the correction is larger than a word substitution. “Sharp” has been removed, but so has the argument that supported it. The original Section 5.1 offered the contrast between a 10 % and a 50 % exposure reduction as evidence of abruptness; those doses are not adjacent, so the contrast cannot speak to the steepness of a boundary, and the argument has been withdrawn. What the data support is weaker and is now stated as such: within the tested grid, retention changes between adjacent configurations in a manner consistent with a threshold, and the location of that change can be bracketed but not resolved. The revision also quotes the grid spacing wherever a boundary is described, so that the reader can judge the resolution: on Qwen, adjacent tested ranks double ($r = 64, 128, 256$, giving effective ranks 29.5, 50.2, 87.6), and on Gemma 3 the relevant interval spans effective ranks 3.07 to 6.37. The phrase used throughout is “a threshold-like transition within the tested grid”, and Section 5.1 now states that “the present experiments do not establish a precise threshold value that will generalise to other settings”. “(Abstract, p. 1; §1, p. 3; §5.1, p. 18; §7, p. 22)”

Author action:

1. Exact permutation and sign-flip tests were added for the rank contrasts, paired t -tests with 95 % confidence intervals for all dose levels, and Hedges’ g for the Gemma 3 anchor contrast;

each is reported with its N and with the pairing assumption stated.

2. Section 3.5 (p. 8) was restructured around the design/observed distinction, and the operational index was qualified at every occurrence.
3. Section 6 (p. 22) gained the per-backbone calibration procedure as the proposed alternative to a universal normalisation, with its cost stated.
4. The contribution statement in Section 1 and the framework discussion in Section 5.1 were rewritten to claim the design and the evidence rather than a measure.
5. Gemma 4 E2B IT is cited as [41], distinguished from Gemma 3n E2B, sourced to the released configuration, and constrained to a supportive role.
6. “Sharp” and “abrupt” were replaced throughout by threshold-like language qualified to the tested grid; the non-adjacent-dose argument for steepness was withdrawn; and grid spacing is now quoted wherever a boundary is described.

Revised manuscript: §1, p. 3; §3.3, p. 6; §3.5, p. 8; §4.1, p. 10; §4.2.1–4.2.3, pp. 12–14; §4.5–4.6, pp. 16–17; §5.1, p. 18; §6, pp. 22–23; §7, p. 22; Table 11, p. 28; Table 13, p. 28; Table S1.

Reviewer #2, Concern #5 — Other format problems:

Reference [12] and [33] share the same arXiv number: arXiv:2408.14572. This is a serious citation error. Reference [32] lists arXiv:2512.15634 with year 2024, which is impossible if the arXiv ID is December 2025. No citation is provided for Gemma 4 E2B IT, despite it being a central backbone. Reference [7] “Keisha et al.” is used heavily but appears to be a preprint with no venue. Its reliability should be clarified. Reference formatting is inconsistent: some entries include conference proceedings, others only arXiv, some lack page numbers or venues.

Author response:

All five are accepted and corrected. The first three were errors of fact in the bibliography; the fourth concerns how a source is characterised in the text; the fifth required an audit of the whole list.

Point raised	Correction
(a) [12] and [33] share arXiv:2408.14572	Two distinct works had been conflated. The identifier belongs to [33] Fawi, <i>CURLoRA: Stable LLM Continual Fine-Tuning and Catastrophic Forgetting Mitigation</i> , arXiv:2408.14572 (2024), which is cited in §2.4 for structural matrix constraints. [12] is Shabgahi et al., <i>ForTIFAI: Fending off Recursive Training Induced Failure for AI Model Collapse</i> , arXiv:2509.08972 (2025), cited in §2.3 for Truncated Cross-Entropy. The identifier now appears exactly once in the bibliography.

(b) [32] year 2024 with a December 2025 arXiv ID	Corrected to 2025. The entry is Rathore et al., <i>How Much Is Too Much? Exploring LoRA Rank Trade-offs for Retaining Knowledge and Domain Robustness</i> , and the venue — ACL-IJCNLP 2025 — is now given alongside the preprint identifier, which resolves the inconsistency the reviewer detected rather than merely changing the digit.
(c) No citation for Gemma 4 E2B IT	Added as [41] Gemma Team, <i>Gemma 4 Technical Report</i> , arXiv:2607.02770 (2026), cited at every point where the backbone is introduced or described. The model is also explicitly distinguished from Gemma 3n E2B in §3.3, and its architectural description is attributed to the released configuration, with the repository identifier and revision hash in supplementary Table S1 (see Concern #4(e)).
(d) [7] is a preprint with no venue	The reviewer is right to ask, and we have made the status visible rather than implicit. The bibliography entry now carries “Non-peer-reviewed preprint” explicitly, and §2.2 marks the comparison as “suggestive context rather than a controlled comparison”, noting that the different dataset, evaluation format and training protocol preclude direct quantitative comparison. We also reduced the load the source carries: no quantitative claim in the paper rests on it, and Shumailov et al. [6] (<i>Nature</i>) is the peer-reviewed anchor for the model-collapse phenomenon. [7] is retained because it is the closest prior work on factual degradation under recursive training and omitting it would misrepresent the literature.
(e) Inconsistent reference formatting	The full list was audited and harmonised to one convention: works published at a venue carry the venue, year and, where applicable, volume and pages; preprints carry the arXiv identifier and year; works that are both carry the venue with the identifier appended, as for [32]. Entries that were previously incomplete were completed against the published records. Three entries not mentioned by the reviewer were also corrected: [31] Soutif-Cormerais et al. (not Biderman et al.), <i>An Empirical Analysis of Forgetting in Pre-trained Models with Incremental Low-Rank Updates</i> , arXiv:2405.18069 (2024); [20] Xu et al., <i>A Probabilistic Perspective on Model Collapse</i> , arXiv:2410.12952 (2025), was missing its arXiv identifier inline; and one further entry had an incomplete venue field. <code>cas-refs.bib</code> is included in the submission package.

Author action:

1. The duplicate arXiv identifier was resolved by separating the two conflated works; each identifier in the bibliography was then checked for uniqueness programmatically.
 2. The year of [32] was corrected and its venue added.
- 41 was added for Gemma 4 and cited at every introduction point.

3. The preprint status of [7] was declared in the entry and the comparison qualified in the text, with [6] added as the peer-reviewed anchor for the phenomenon.
4. Every entry was checked against the published record for venue, year, volume, pages and identifier, and the list was harmonised to a single convention. Three further entries ([31], [20], and one with an incomplete venue) not mentioned by the reviewer were found and corrected in the same pass.

*Revised manuscript: **cas-refs.bib** (full list); entries [6], [7], [12], [20], [31], [32], [33], [41]; §2.2, p. 4; §2.3–2.4, p. 4; §3.3, p. 6; Table S1.*